

Redwan Ahmed Rizvee

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RESEARCH INTERESTS

My research bridges **Database Systems, Visual Analytics, and Human Perceptual Cognition**. I develop perception-aware sampling operators, Approximate Query Processing (AQP) algorithms, and density estimation models that preserve critical topological outliers and local structures in massive visual datasets and foundation model latent spaces. Developed an interactive, web-based demonstration of the perception-aware sampling framework (**Live Demo: paws**), showcasing real-time visual fidelity across diverse topologies.

EDUCATION

University of Massachusetts Amherst

PhD in Computer Science (Advisor: Dr. Alexandra Meliou)

Amherst, MA

Sep. 2025 – Present

- **Research Focus:** Formulating perception-aware sampling algorithms and Approximate Query Processing (AQP) execution operators to optimize visual fidelity and query latency over massive data viewpoints.
- Addressing representational fidelity and system scalability bottlenecks at the intersection of cognitive perceptual bounds and database runtime engines.

University of Dhaka

Master of Science in Computer Science and Engineering

Dhaka, Bangladesh

Jan. 2019 – Sep. 2021

- CGPA: 4.00/4.00 (Ranked 1st in class)

University of Dhaka

Bachelor of Science in Computer Science and Engineering

Dhaka, Bangladesh

Jan. 2015 – Dec. 2018

- CGPA: 3.85/4.00 (Ranked 1st in class) | Dean's Award, Gold Medalist

SELECTED PUBLICATIONS

- Z. Moumoulidou, H. Elhamdadi, **R. A. Rizvee**, K. Yang, S. Mitra, C. Xiong Bearfield, A. Meliou, "Perception-Augmented Databases for Effective Visual Analytics", in *ACM SIGMOD International Conference on Management of Data (SIGMOD)*, 2027 [**Under Revision**].
- **R. A. Rizvee**, C. F. Ahmed, M. F. Arefin, C. K. Leung, "A new tree-based approach to mine sequential patterns," *Expert Systems with Applications*, vol. 242, 122754, 2024.
- **R. A. Rizvee**, R. Hassan, M. M. Khan, "Reinforcement Learning-Based Formulations With Hamiltonian-Inspired Loss Functions for Combinatorial Optimization," *IEEE Access*, 2024.
- **R. A. Rizvee**, et al., "LeafNet: A proficient convolutional neural network for detecting seven prominent mango leaf diseases," *Journal of Agriculture and Food Research*, 2023.

TECHNICAL SKILLS

- **Data Systems & Processing:** Approximate Query Processing (AQP), Vectorized Batch Processing, Statistical Data Sampling & Synopses, Relational Databases (PostgreSQL, MySQL), SQL.
- **Languages:** Python, C++, C, Java, SQL, JavaScript (ES6+), Shell Scripting.
- **Machine Learning & AI Platforms:** PyTorch, Computer Vision (OpenCV), Saliency Modeling, Attention Mechanisms, Transformer Architectures, Scikit-learn.
- **Visual Computing & Analytics:** Perception-Aware Sampling, Visual Analytics, Density Estimation, Interactive Dashboards.
- **Tools & Environments:** Linux/Unix, Git, Bash/Shell, Flask, Django, React, L^AT_EX.

RESEARCH & PROFESSIONAL EXPERIENCE

University of Massachusetts Amherst (DREAM Lab)

Graduate Research Assistant

Amherst, MA

Sep. 2025 – Present

- Co-authored research under revision at **ACM SIGMOD 2027**, designing perception-aware sampling algorithms that preserve local density variations and critical outliers in visual data representations.
- Developed a lightweight, density-based proxy model to dynamically tune sampling hyper parameters, substantially cutting computational runtime while guaranteeing visual and structural fidelity.
- Evaluated sampling quality across diverse synthetic and empirical distributions, benchmarking perceptual error and outlier retention against standard sampling baselines.
- Maintained and enhanced an interactive web-based demonstration platform (**paws**) showcasing comparative sampling behavior and real-time visual fidelity across topologies.

TigerIT Bangladesh Limited
Research Scientist

Dhaka, Bangladesh
 July 2020 – Feb. 2022

- Designed and deployed Encoder-Decoder neural network architectures equipped with Attention mechanisms for audio processing and natural language modeling tasks.
- Optimized deep learning inference workflows and acoustic data processing pipelines, reducing latency and enhancing accuracy in production deployment.
- Led research initiatives in statistical sensor data processing and implemented robust signal-analysis algorithms for commercial environments.

Cognitive Agents and Interaction Lab (CAIL)
Co-Investigator

Dhaka, Bangladesh
 Jan. 2023 – Aug. 2025

- Supervised research projects applying Deep Learning and Reinforcement Learning to agent-based decision-making and human-in-the-loop systems.
- Architected COLAS (Coding Lab Assistant), an automated source code evaluation platform integrating LLM reasoning, code analysis, and plagiarism detection.

Data Mining Research Lab
Research Assistant

Dhaka, Bangladesh
 Jan. 2018 – Jul. 2025

- Developed scalable, resource-efficient sequential pattern mining algorithms tailored for complex time-series datasets.
- Designed mathematical metrics for high-dimensional correlation analysis, clustering, and anomaly detection.

Data and Design Lab (DnD Lab)
Research Assistant

Dhaka, Bangladesh
 Apr. 2019 – Sep. 2019

- Collaborated with BRAC on the Bangladesh Quality of Life (BDQOL) indicator framework, designing data processing pipelines and interactive visualization dashboards for high-dimensional socio-economic data.

POSTERS & PRESENTATIONS

New England Database Day (NEDB), 2026 | *[Poster]*
Poster Presenter

Jan. 2026

Presented research on perception-aware sampling techniques for large-scale data visualization to the regional database and systems research community.

LLMs as Visual Survey Agents for Scatter Plot Similarity | *[Poster]*
PhD Mentor (First Author: Bach Nguyen)

2026

Investigated multimodal foundation models (Claude, GPT, Gemini) as visual survey agents, quantifying position bias and opacity sensitivity in scatter plot similarity tasks.

HONORS & AWARDS

- **Research Excellence Recognition** (2025): Awarded by the University of Dhaka for exemplary scholarly contributions.
- **Dean's Award & Gold Medal** (2019): Conferred for achieving the highest academic distinction across the Faculty of Engineering.
- **Fellowship Scholarship**: Awarded by the Information and Communication Technology (ICT) Division, Bangladesh, for high-impact research.